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50 Most Asked Azure Interview Questions



Note: These abbreviations indicate companies where the interview question is commonly asked.

TC = TCS

INF = Infosys

HX = Hexaware

PST = Persistent

CG = CGI

CGZ = Cognizant

CGM = Capgemini

WP = Wipro

MP = Mphasis



Q.1 What is Azure App Service and how do you deploy a .NET application to it? (TC, INF, HX, PST, CG)

Azure App Service is a managed web hosting service where you can deploy .NET applications using Visual Studio, GitHub, or Azure DevOps.

Q.2 What are the different options to deploy a .NET Web API on Azure? (TC, INF, WP)

You can deploy a .NET Web API to App Service, AKS, Azure Container Apps, or Azure Virtual Machines.

Q.3 Why did you choose Azure App Service instead of AKS or Azure VM? (HX, PST)

App Service is easier to manage because Azure handles the infrastructure.

Q.4 What is the difference between Azure App Service and Azure Virtual Machine? (TC, INF, CG)

App Service is fully managed, while a VM gives you full control over the operating system.



Q.5 What is Azure Functions and when would you use it? (HX, INF, CGZ, WP)

Azure Functions is a serverless compute service used for event-driven or background tasks.

Q.6 How do you secure Azure Functions using Azure AD and JWT? (HX, CGZ, PST)

In function app in Authentication add identity provider, configure secret keys and apply Oauth/OpenId flow.

Q.7 How do you implement Role-Based Access Control (RBAC) in Azure Functions? (HX, PST)

RBAC is implemented by assigning Azure roles to users or managed identities.

Q.8 How do you handle JWT token expiration and refresh tokens in Azure? (HX, WP)

Use refresh tokens to obtain a new access token after expiration.



Q.9 What are Durable Functions and when should they be used? (HX, TC)

Durable Functions are used for long-running workflows and orchestration.

Q.10 How do you monitor Azure Functions and receive alerts when they fail? (TC, MP)

Use Application Insights and Azure Monitor alerts.

Q.11 What is Azure Storage Account and what are its different storage types? (TC, INF)

A Storage Account stores Azure data like Blobs, Files, Queues, and Tables.

Q.12 How do you secure an Azure Storage Account from unauthorized access? (PST)

Use RBAC, Private Endpoints, firewalls, and Managed Identity.



**Q.13 What is Azure Key Vault and why is it important?
(TC, INF, PST, CGM)**

Azure Key Vault securely stores secrets, keys, and certificates.

Q.14 How do you connect an ASP.NET Core application to Azure Key Vault? (INF, PST)

Use the Azure Key Vault configuration provider with Managed Identity.

Q.15 What is Managed Identity and how does it work with Azure Key Vault? (HX, INF, PST)

Managed Identity lets Azure services securely access Key Vault without storing credentials.

Q.16 How do you implement CI/CD using Azure DevOps? (TC, INF, CGM)

Create build and release pipelines to automate testing and deployment.



**Q.17 Explain the stages of an Azure DevOps pipeline.
(TC, INF)**

The stages are Build, Test, Release, and Deploy.

Q.18 How do you deploy Docker containers to Azure App Service? (HX)

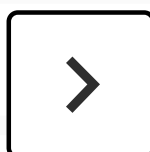
Push the container to ACR and configure App Service to use it.

Q.19 What is Azure Container Registry (ACR)? (HX)

ACR is Azure's private Docker image registry.

Q.20 What is Azure Kubernetes Service (AKS) and when would you use it? (HX, TC)

AKS is a managed Kubernetes service for containerized applications.



Q.21 How do you configure autoscaling in Azure App Service? (HX)

Configure scaling rules based on metrics like CPU or memory.

Q.22 What is Azure Service Bus? (TC, CGZ, CG)

Azure Service Bus is a reliable message broker for asynchronous communication.

Q.23 What is the difference between Queue and Topic in Azure Service Bus? (CG, CGZ)

Queue is one-to-one messaging, while Topic is one-to-many messaging.

Q.24 When would you choose Azure Service Bus over REST API communication? (CG, CGZ)

Use Service Bus for reliable asynchronous communication.



Q.25 What is Azure Event Grid? (TC, PST)

Event Grid is an event routing service for reacting to Azure events.

Q.26 What is Azure Event Hub? (TC, PST)

Event Hub is a high-throughput event ingestion service.

Q.27 What is the difference between Azure Event Grid and Azure Event Hub? (TC, PST)

Event Grid routes events, while Event Hub streams large volumes of events.

Q.28 What is Azure API Management (APIM) and why is it used? (TC, CGM)

APIM secures, publishes, and manages APIs.



Q.29 How do you secure an ASP.NET Core Web API hosted in Azure? (HX, CGZ)

Use Azure AD authentication, HTTPS and authorization policies.

Q.30 How do you authenticate Azure APIs using Azure AD? (HX, CGZ)

Register the API in Azure AD and validate access tokens.

Q.31 What is Azure Monitor? (HX, MP)

Azure Monitor collects and analyzes metrics and logs.

Q.32 What is Application Insights? (HX, INF, MP)

Application Insights monitors application performance and failures.



Q.33 How do you configure centralized logging using Application Insights? (INF, MP)

Enable Application Insights and send logs from all applications.

Q.34 How do you write KQL (Kusto Query Language) queries for troubleshooting? (HX)

Use KQL to search and analyze Azure Monitor logs.

Q.35 How do you troubleshoot performance issues in Azure App Service? (HX, INF)

Analyze CPU, memory, logs, and Application Insights telemetry.

Q.36 How do you implement distributed caching in Azure? (TC)

Use Azure Redis Cache as a shared cache.



Q.37 What is Azure Redis Cache and when would you use it? (TC)

Azure Redis Cache stores frequently accessed data in memory.

Q.38 How do you store secrets without using appsettings.json? (PST, INF)

Store secrets in Azure Key Vault.

Q.39 What are Private Endpoints and VNet Integration? (PST)

They provide secure private network connectivity to Azure services.

Q.40 How do you secure communication between Azure services? (PST)

Use Managed Identity, Private Endpoints and HTTPS.



Q.41 What are deployment slots in Azure App Service? (CGM, TC)

Deployment slots allow testing before swapping to production.

Q.42 What is Blue-Green Deployment in Azure? (CGM)

Blue-Green deployment switches traffic between two identical environments.

Q.43 What is Canary Deployment in Azure? (CGM)

Canary deployment releases changes gradually to a small group of users.

Q.44 How do you roll back a failed deployment in Azure? (CGM)

Swap back the deployment slot or redeploy the previous version.



**Q.45 What Azure AI services have you worked with?
(TC, INF)**

Azure AI includes services like Azure OpenAI, Document Intelligence, and Vision.

Q.46 How do you integrate Azure OpenAI or Azure AI Services into a .NET application? (TC)

Use the Azure SDK or REST APIs with authentication.

Q.47 How do you optimize Azure cloud costs?

Right-size resources, use autoscaling, and monitor usage.

Q.48 How do you design a highly available Azure architecture?

Use multiple availability zones, load balancers, and redundant services.



Q.49 How do you implement disaster recovery in Azure?

Use Azure Site Recovery, backups, and geo-redundant storage.

Q.50 Describe the Azure architecture used in your current project.

Explain the Azure services, data flow, security, and deployment used in your application.



The FREE Azure Webinar link is in the comment

